

GEOLOGICAL FEATURES

Most of Earth's geological features are associated with plate boundaries. At constructive (or divergent) boundaries, plates move apart and new crust is added. Examples are midocean ridges and the East African Rift. At destructive (or convergent) boundaries, two plates push against each other, producing a range of features, depending on

the nature of the crust on each plate. Many plate boundaries are associated with an increased frequency of volcanism, earthquakes, or both.

THE SAN ANDREAS FAULT

This fault in California, known for producing earthquakes, marks a transform boundary where two plates push past each other.

AFRICA *east* and ASIA *southwest*

East African Rift

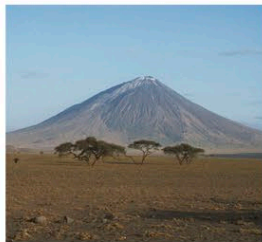


LOCATION Extending from Mozambique northward through East Africa, the Red Sea, and into Lebanon

TYPE Series of rift faults

LENGTH 5,300 miles (8,500 km)

The East African Rift provides an example of the geological process of rifting—the stretching and tearing apart of a section of continental crust by a plume of hot magma pushing up



OL DOINYO LENGAI

This active volcano in northern Tanzania sits in the middle of the east African part of the East African Rift.

underneath it. Rifting is associated with the development of a constructive plate boundary, which is formed as ascending magma creates new crust and pushes the plates on either side of the rift apart. The main section of the East African Rift runs (in two branches) through eastern Africa. Over tens of millions of years, rifting in this region has caused extensive faulting, the collapse of large chunks of crust, and associated features such as volcanism and a series of lakes in the subsided sections. As rifting continues, it is anticipated that a large area of eastern

Africa will eventually split off as a separate island. A northern arm of the rift valley runs up the Red Sea and eventually reaches Lebanon, in the north. This coincides with a divergent boundary that is pushing Arabia away from Africa.



THE NORTHERN RED SEA

The Gulf of Aqaba (center right), a branch of the Red Sea, forms part of the northern arm of the East African Rift. The Gulf of Suez (center) is a side branch of the rift.

ATLANTIC OCEAN

Mid-Atlantic Ridge



LOCATION Extending from the Arctic Ocean to the Southern Ocean

TYPE Slow-spreading midocean ridge

LENGTH 10,000 miles (16,000 km)

The Mid-Atlantic Ridge is the longest mountain chain on Earth and one of its most active volcanic regions, albeit mainly underwater. The ridge sits on top of the Mid-Atlantic Rise, a bulge that runs the length of the Atlantic Ocean floor. Both rise and ridge coincide with plate boundaries that divide the North and South American plates, on the west, from the Eurasian and African plates, on the east. These are constructive plate boundaries, where new ocean crust is formed by magma upwelling from

Earth's mantle. As this crust forms, the plates on either side are pushed away from the ridge at a rate of 0.4–4 in (1–10 cm) a year, widening the Atlantic

basin. The discovery in the 1960s of this spreading of the Atlantic sea floor—evidenced by the fact that crustal material near the ridge is younger than that farther away—led to general acceptance of the theory of continental drift. The ridge is a site of extensive earthquake activity and volcanism, along with many seamounts (isolated underwater mountains). Where the volcanoes break the ocean surface, they have formed islands such as Iceland and the Azores.



SURTSEY

Between 1963 and 1967, a massive and dramatic submarine eruption, from a section of the Mid-Atlantic Ridge to the south of Iceland, produced the new island of Surtsey.



BLACK SMOKERS

Hydrothermal vents are underwater geysers located near midocean ridges. The hot water spewed out by some vents, called "black smokers," is discolored by the dark mineral iron sulfide.

